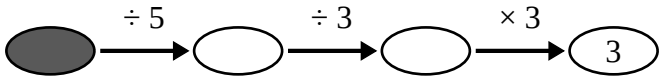
Preview: y2007_q05 (v2)

Working backwards: undo each operation, last one first

Part A: Each question is worth 3 points

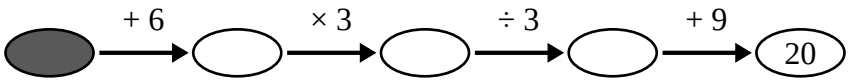
1. Emma wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 3.



What is Emma's favourite number?

A. 3 ***B.*** 14 ***C.*** 15 ***D.*** 16 ***E.*** 18

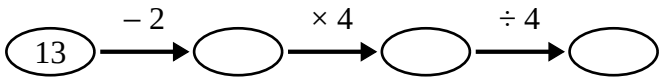
2. Nina wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 20.



What is Nina's favourite number?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 20 ***E.*** 35

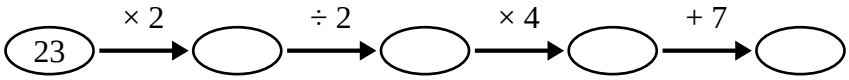
3. Follow the arrows and do each calculation in turn.



Which number comes out at the end?

A. 12 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 8

4. Follow the arrows and do each calculation in turn.



Which number comes out at the end?

A. 100 ***B.*** 99 ***C.*** 98 ***D.*** 82 ***E.*** 23

Part B: Each question is worth 4 points

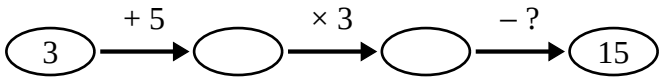
5. Sara thinks of a number, multiplies by 3, then takes away 1, and gets 32.
Which number did Sara think of?

A. 32 ***B.*** 13 ***C.*** 11 ***D.*** 10 ***E.*** 9

6. Leo thinks of a number, adds 2, then multiplies by 2, then takes away 4, and gets 58.
Which number did Leo think of?

A. 26 ***B.*** 29 ***C.*** 32 ***D.*** 58 ***E.*** 116

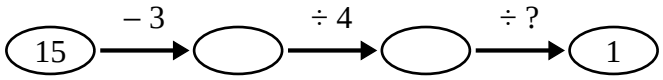
7. The arrows show calculations done one after the other. One number is hidden.



Which number is hidden behind the question mark?

A. 11 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 6

8. The arrows show calculations done one after the other. One number is hidden.

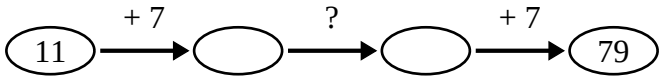


Which number is hidden behind the question mark?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

Part C: Each question is worth 5 points

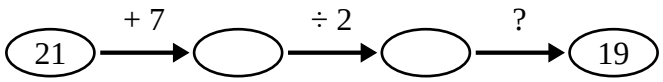
9. The arrows show calculations done one after the other, but one arrow has lost its calculation: it shows a question mark.



Which calculation belongs on that arrow?

A. – 4 ***B.*** ÷ 3 ***C.*** × 4 ***D.*** + 9 ***E.*** + 4

10. The arrows show calculations done one after the other, but one arrow has lost its calculation: it shows a question mark.



Which calculation belongs on that arrow?

A. × 5 ***B.*** – 5 ***C.*** – 2 ***D.*** – 1 ***E.*** + 5

Answer Key

1. **C**	**2. A**	**3. B**	**4. B**	**5. C**	**6. B**
7. **B**	**8. C**	**9. C**	**10. E**		

Solutions

1. C (15) — Go backwards and do the opposite: 3 ÷ 3 = 1, 1 × 3 = 3, 3 × 5 = 15. [2007 Q5 · favourite_number]

2. A (5) — Go backwards and do the opposite: 20 – 9 = 11, 11 × 3 = 33, 33 ÷ 3 = 11, 11 – 6 = 5. [2007 Q5 · favourite_number]

3. B (11) — 13 – 2 = 11, 11 × 4 = 44, 44 ÷ 4 = 11. [2007 Q5 · forwards]

4. B (99) — 23 × 2 = 46, 46 ÷ 2 = 23, 23 × 4 = 92, 92 + 7 = 99. [2007 Q5 · forwards]

5. C (11) — Go backwards: 32 + 1 = 33, 33 ÷ 3 = 11. [2007 Q5 · think_of_a_number]

6. B (29) — Go backwards: 58 + 4 = 62, 62 ÷ 2 = 31, 31 – 2 = 29. [2007 Q5 · think_of_a_number]

7. B (9) — Forwards from 3 we reach 24 before the hidden step; backwards from 15 we reach 15 after it. 24 – 9 = 15. [2007 Q5 · missing_number]

8. C (3) — Forwards from 15 we reach 3 before the hidden step; backwards from 1 we reach 1 after it. 3 ÷ 3 = 1. [2007 Q5 · missing_number]

9. C (× 4) — Before that arrow we have 18; working back from 79, after it we need 72. Only × 4 turns 18 into 72. [2007 Q5 · which_operation]

10. E (+ 5) — Before that arrow we have 14; working back from 19, after it we need 19. Only + 5 turns 14 into 19. [2007 Q5 · which_operation]